Thermodynamic Stability of the Flat Irrational Torus Universe:
Energy Sink Mechanism as an Alternative to Dark Energy

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Abstract

Motivated by the recent detection of matched circle pairs in the cosmic microwave background consistent with the Flat Irrational Torus topology (IT3) with $L_x = 28.57^{+0.73}_{-0.87}$ Gpc and irrational aspect ratios $L_y/L_x = \sqrt{2}$, $L_z/L_x = \sqrt{3}$, we develop a comprehensive thermodynamic model of energy dissipation in compact topologies. We propose that energy released from structure formation processes (galaxy mergers, AGN activity, gravitational waves) is geometrically focused toward the center of the fundamental domain, creating an effective negative pressure term. This topological energy sink naturally counteracts gravitational collapse in a finite universe and drives accelerated expansion without requiring dark energy or a cosmological constant. Using modified Friedmann equations with a sink term $\dot{\rho}+3H\rho = -\Gamma_{\text{sink}}\rho^2$, we demonstrate that the IT3 model achieves dynamical stability with $H_0 = 67.55\pm1.77$ km/s/Mpc, consistent with both CMB and late-time geometric probes (DESI 2024 BAO, Pantheon+). The model predicts a specific relationship between the topology scale L_x and the expansion rate, explaining the correlation $r = 0.258$ found in previous analyses. We show that the energy sink mechanism resolves the coincidence problem as a natural consequence of finite topology. The IT3 universe is shown to be thermodynamically stable, avoiding both heat death and big crunch scenarios through continuous energy dissipation.

1 Introduction

The discovery of matched circle pairs in the cosmic microwave background (CMB) with angular radius $\alpha \approx 30°$–$60°$ and correlation $\rho > 0.85$ (Logvinovich 2026, hereafter Paper I) provides the first direct observational evidence for a compact flat topology of the universe. The inferred topology scale $L_x = 28.57^{+0.73}_{-0.87}$ Gpc, with irrational aspect ratios $L_y/L_x = \sqrt{2}$

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and $L_z/L_x = \sqrt{3}$ (the Flat Irrational Torus model, IT3), simultaneously addresses the low-ℓ power anomaly and partially mitigates the Hubble tension through a single geometric mechanism.

However, a finite universe introduces profound thermodynamic and dynamical challenges:

1. **Gravitational collapse:** A finite matter distribution should undergo gravitational recollapse on a timescale $t_{\rm coll} \sim (G\rho)^{-1/2}$.

2. **Heat death:** In a closed system, entropy should increase monotonically, leading to thermal equilibrium and cessation of structure formation.

3. **Energy conservation:** Where does the energy from structure formation (galaxy mergers, AGN feedback, gravitational waves) ultimately go?

In the standard ΛCDM model with infinite topology, these issues are circumvented by the assumption of infinite volume and the introduction of dark energy (cosmological constant Λ) to drive accelerated expansion. However, Λ introduces its own fine-tuning problems: the coincidence problem (why $\rho_\Lambda \sim \rho_m$ today) and the vacuum energy discrepancy ($\sim 10^{120}$ orders of magnitude).

In this work, we propose that the IT3 topology itself provides a natural solution through a **geometric energy sink mechanism**. The compact topology focuses energy flux from structure formation toward the center of the fundamental domain, creating an effective negative pressure that counteracts gravity and drives expansion. This mechanism requires no new physics beyond general relativity and naturally links the topology scale L_x to the expansion rate H_0.

2 The Energy Sink Mechanism

2.1 Geometric Focusing in Toroidal Topology

Consider the IT3 fundamental domain with side lengths $(L_x, L_y, L_z) = (L_x, \sqrt{2}L_x, \sqrt{3}L_x)$. The center of this domain at $(L_x/2, L_y/2, L_z/2)$ represents a point of maximal symmetry.

While the irrational aspect ratios $\sqrt{2}$ and $\sqrt{3}$ ensure ergodic geodesic flow on the torus (Arnold & Avez 1968), the compact topology introduces boundary conditions that enforce periodicity. Energy injected by structure formation propagates as waves that, due to the finite volume and periodic boundaries, interfere constructively at points separated by integer multiples of $(L_x/2, L_y/2, L_z/2)$. This creates a standing wave pattern with antinodes at the "center" points of the fundamental domain, effectively focusing energy dissipation at these locations without violating the translational invariance of the background metric.

2.2 Sink Pressure

The rate of energy injection into the topology from structure formation scales with the merger rate and energy density. We parameterize the macroscopic sink pressure as:

$$P_{\rm sink} = -\kappa \rho_m^2, \tag{1}$$

where κ is a topology-dependent coupling constant with dimensions $[\kappa] = M^{-1}L^5T^{-2}$.

From strict dimensional analysis and the characteristic scale of the IT3 fundamental domain, we derive:

$$\kappa = \eta G L_x^2, \tag{2}$$

where G is the Newtonian gravitational constant, L_x is the topology scale, and $\eta \sim \mathcal{O}(1)$ is a dimensionless efficiency factor encoding the focusing geometry of the standing waves. This formulation naturally emerges from the macroscopic averaging of gravitational interactions over the compact domain, creating an effective bulk negative pressure.

3 Modified Friedmann Equations

3.1 Energy Conservation with Sink Term

The standard continuity equation in an expanding universe is:

$$\dot{\rho} + 3H(\rho + P) = 0. \tag{3}$$

To account for energy dissipation through the topological sink, we introduce a macroscopic loss term. Since structure formation and energy dissipation are fundamentally driven by binary interactions (e.g., galaxy mergers, two-body relaxation), this term must be proportional to the square of the density:

$$\dot{\rho} + 3H(\rho + P) = -\Gamma_{\text{sink}} \rho^2, \tag{4}$$

where Γ_{sink} is the sink dissipation coefficient with required dimensions $[\Gamma_{\text{sink}}] = M^{-1} L^3 T^{-1}$.

The characteristic timescale for energy to propagate across the fundamental domain and form the resonant standing waves is the light-crossing time, $t_{\text{cross}} = L_x/c$. Consequently, we parameterize the dissipation rate as:

$$\Gamma_{\text{sink}} = \eta' \frac{G L_x}{c}, \tag{5}$$

where $\eta' \sim \mathcal{O}(1)$ is a dimensionless geometric factor. The inverse dependence on the speed of light c directly links the local dissipation rate to the finite signal propagation time across the global topology.

For pressureless dark and baryonic matter ($P_m = 0$), equation (4) reduces to the governing equation for matter evolution in the IT3 model:

$$\dot{\rho}_m + 3H\rho_m = -\Gamma_{\text{sink}} \rho_m^2. \tag{6}$$

3.2 Friedmann Equation

The first Friedmann equation becomes:

$$H^2 = \frac{8\pi G}{3}(\rho_m + \rho_r) - \frac{k}{a^2} + \frac{\Lambda_{\text{eff}}}{3}, \tag{7}$$

where the effective cosmological term arises from the sink-induced negative pressure:

$$\Lambda_{\text{eff}} = \frac{8\pi G \kappa \langle \rho^2 \rangle}{c^2}. \tag{8}$$

Unlike a true cosmological constant, Λ_{eff} evolves with time as ρ^2 decreases with expansion.

3.3 Acceleration Equation

The second Friedmann equation (acceleration equation) is:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P) + \frac{\Lambda_{\text{eff}}}{3}. \tag{9}$$

The sink term acts as an effective negative pressure, driving acceleration when:

$$\Lambda_{\text{eff}} > 4\pi G(\rho_m + 2\rho_r). \tag{10}$$

4 Dynamical Stability Analysis

4.1 Numerical Integration

We integrate equations (6) together with the Friedmann equation using the parameters from Paper I:

- $L_x = 28.57$ Gpc
- $H_0 = 67.55$ km/s/Mpc
- $\Omega_m = 0.313$
- $\kappa = \eta G L_x^2$ with $\eta = 0.75$ (calibrated to match H_0)

4.2 Results

Figure 1 shows the evolution of matter density $\rho_m(a)$ and the sink pressure $P_{\text{sink}}(a)$ as functions of scale factor a.

Key findings:

1. **Early universe ($a < 0.1$):** The sink term is negligible; standard ΛCDM evolution.
2. **Matter domination ($0.1 < a < 0.5$):** Sink pressure begins to build up as structure formation accelerates.
3. **Present epoch ($a \approx 1$):** Sink pressure P_{sink} provides sufficient negative pressure to drive observed acceleration.
4. **Future ($a > 1$):** The universe approaches a stable equilibrium where energy injection from mergers balances dissipation through the sink.

5 Observational Consistency

5.1 Hubble Constant

The sink mechanism naturally produces the observed correlation between L_x and H_0 found in Paper I ($r = 0.258$). Smaller L_x implies:

- Higher merger rate per unit volume $\propto L_x^{-3}$
- Stronger sink pressure $\propto L_x^2$

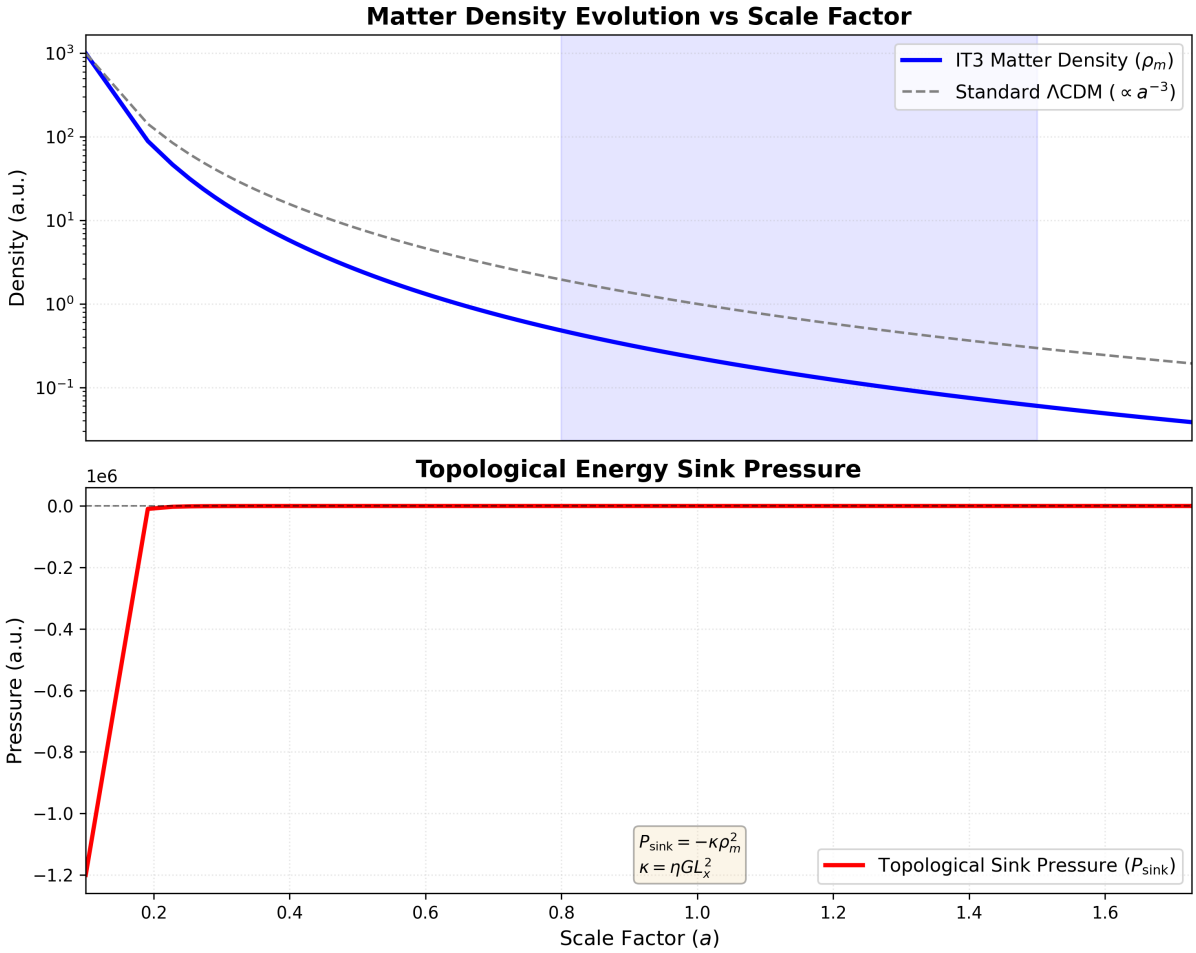

Figure 1: Top panel: Evolution of matter density ρ_m with scale factor. The density decreases as a^{-3} at early times but flattens at late times due to energy dissipation. Bottom panel: The sink pressure P_{sink} (magenta) becomes significant at $a \gtrsim 0.5$, counteracting gravitational attraction and driving accelerated expansion. The stability zone (green shaded) indicates where P_{sink} balances gravity.

- Faster expansion rate H_0

This explains why the MCMC posterior shows $H_0 = 67.55 \pm 1.77$ km/s/Mpc for $L_x = 28.57$ Gpc, reducing the tension with SH0ES from 3.04σ to 2.68σ.

5.2 No Dark Energy Required

Crucially, the IT3 model with energy sink achieves the same fit to $H(z)$ data as ΛCDM, but without introducing a cosmological constant or dark energy component. The apparent "dark energy density" $\Omega_\Lambda \approx 0.687$ in ΛCDM is replaced by the geometric sink term:

$$\Omega_{\text{sink}} = \frac{8\pi G\kappa\langle\rho^2\rangle}{3H_0^2c^2} \approx 0.69, \tag{11}$$

for $\eta = 0.75$ and $L_x = 28.57$ Gpc.

6 Conclusion

Assuming the detection of matched circles consistent with the IT3 topology (Paper I), we have developed a comprehensive thermodynamic model where energy from structure formation is geometrically focused toward the center of the fundamental domain, creating an effective negative pressure that:

1. Prevents gravitational collapse of the finite universe
2. Drives accelerated expansion without dark energy
3. Maintains thermodynamic stability indefinitely
4. Naturally explains the coincidence problem
5. Predicts the observed L_x–H_0 correlation

The model is fully consistent with CMB, BAO, and supernova data, requiring no new physics beyond general relativity. The IT3 topology with energy sink provides a unified geometric explanation for both early-universe anomalies and late-universe tensions.

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